

Week 5 Assignment

Exercise 1:

Mahendra Singh Dhoni has put up commendable achievements in his bating career with a boasting batting average of 51.08 (ODIs) & (38.09) in Tests and a striking rate of 89.27 (ODIs) & 59.11 (Tests). In a span of 10-11 years, not only has he become India's most successful captain, but he's also changed the limited-overs game in ways that one could only dream of. If his entire Test bating statistics are referred, the runs scored by Dhoni in the continual matches followed a particular pattern of series. The series looks like 0, 6, 24, 60, 120, 210, 336, Predict the runs scored by Dhoni in this pattern of series up to "N" matches.

namespace Exercise1

```
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int n = int.Parse(Console.ReadLine());
            Console.WriteLine("Runs");
            for (int i = 1; i <= n; i++)
            {
                Console.Write(i*(i-1)*(i+1)+" ");
            }
        }
    }
}
```

```
8
Runs
0 6 24 60 120 210 336 504
C:\Training\.Net\Exercise1\bin\Debug\net8.0\Exercise1.exe (process 5980) exited with code 0 (0x0).
Press any key to close this window . . .
```

Exercise 2:

Mahi has learnt about the different properties of circles like radius, centre, diameter, circumference ... To make the tutorial class on circles more interesting, her teacher organises a game. The players will be given 2 circles --- One circle A with radius r_a and with central coordinate (x_a, y_a) and second circle B with radius r_b and with central coordinate (x_b, y_b) . The player needs to say

- B is in A
- A is in B
- A and B intersect
- A and B do not intersect

You may assume that A and B are not identical. Can you please help Mahi in solving this game?

```

namespace Exercise2
{
    internal class Program
    {
        static void Main(string[] args)
        {
            string[] a = Console.ReadLine().Split();
            double ax = double.Parse(a[0]);
            double ay = double.Parse(a[1]);
            double ar = double.Parse(a[2]);
            string[] b = Console.ReadLine().Split();
            double bx = double.Parse(b[0]);
            double by = double.Parse(b[1]);
            double br = double.Parse(b[2]);
            double d = Math.Sqrt(Math.Pow(ax - bx, 2) + Math.Pow(ay - by, 2));

            if (d + br < ar)
            {
                Console.WriteLine("B is in A");
            }
            else if (d + ar < br)
            {
                Console.WriteLine("A is in B");
            }
            else if (d <= ar + br && d >= Math.Abs(ar - br))
            {
                Console.WriteLine("A and B intersect");
            }
            else
            {
                Console.WriteLine("A and B do not intersect");
            }
        }
    }
}

```

```

0 5 7
3 4 4
A and B intersect
C:\Training\.Net\Exercise2\bin\Debug\net8.0\Exercise2.exe (process 24660) exited with code 0 (0x0).
Press any key to close this window . . .

```

Exercise: 3

An organization wants to calculate an employee's Net Salary based on:

Basic Salary HRA = 20% of Basic

DA = 10% of Basic

PF = 12% of Basic

If Basic Salary < 15,000 → No PF deduction.

Create a Class Library named SalaryCalculator and create a method in library as signature below:

public static double CalculateNetSalary(double basicSalary)

Consume this library in a console application. Get proper details of an individual Employee and display proper result. Implement Basic Exception Handling also in Library.

Library

namespace SalaryCalculator

```
{
    public class Class1
    {
        public static double CalculateNetSalary(double basicSalary)
        {
            try
            {
                if (basicSalary <= 0)
                {
                    throw new ArgumentException("Salary must be greater than Zero.");
                }

                double hra = 0.20 * basicSalary;
                double da = 0.10 * basicSalary;
                double pf = 0;
                if (basicSalary >= 15000)
                {
                    pf = 0.12 * basicSalary;
                }

                double ns = basicSalary + hra + da - pf;
                return ns;
            }
            catch (Exception e)
            {
                throw new Exception("Error in Salary Calculation: " + e.Message);
            }
        }
    }
}
```

Console App

using SalaryCalculator;

namespace Exercise3

```
{
    internal class Program
```

```

{
    static void Main(string[] args)
    {
        try
        {
            Console.Write("Enter name:");
            string name = Console.ReadLine();
            Console.Write("Enter id");
            int empId = int.Parse(Console.ReadLine());
            Console.Write("Basic Salary: ");
            double basicSalary = double.Parse(Console.ReadLine());
            double netSalary = Class1.CalculateNetSalary(basicSalary);
            Console.WriteLine("Employee ID : " + empId);
            Console.WriteLine("Employee Name : " + name);
            Console.WriteLine("Basic Salary : " + basicSalary);
            Console.WriteLine("Net Salary : " + netSalary);
        }
        catch (FormatException)
        {
            Console.WriteLine("Invalid input format.");
        }
        catch (Exception ex)
        {
            Console.WriteLine(ex.Message);
        }
    }
}

```

```

Enter name:Rohith
Enter id103
Basic Salary: 40000
Employee ID : 103
Employee Name : Rohith
Basic Salary : 40000
Net Salary : 47200

C:\Training\.Net\Exercise3\bin\Debug\net8.0\Exercise3.exe (process 18784) exited with code 0 (0x0).
Press any key to close this window . . .

```

Exercise 4:

Create an application to produce an Electricity bill using console application with proper format.

Draw Boxes, lines and columns using characters only.

For input get CustomerId, CustomerName, Address, PhoneNumber, EmailId, Type of connection (Industrial, Business, Domestic, Agricultural), Previous Reading, Current Reading.

Calculate Unit Consumed and decide Electricity charges on slab basis as:

First 100 units -> ₹1.5 / Unit consumed

Next 150 units -> ₹2.5 / Unit consumed

Next 300 units -> ₹4.5 / Unit consumed

Above 1000 Units -> ₹7.5 / Unit consumed

Meter Rent based on Category should be charged as fixed amount.

Industrial -> 2500

Business -> 1500

Domestic -> 1000

Agricultural -> Free

namespace Exercise4

```
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.Write("Customer ID : ");
            int custId = int.Parse(Console.ReadLine());

            Console.Write("Customer Name : ");
            string name = Console.ReadLine();

            Console.Write("Address : ");
            string addr = Console.ReadLine();

            Console.Write("Phone Number : ");
            string phone = Console.ReadLine();

            Console.Write("Email ID : ");
            string email = Console.ReadLine();

            Console.Write("Connection Type : ");
            string type = Console.ReadLine().ToLower();

            Console.Write("Previous Reading : ");
            int prev = int.Parse(Console.ReadLine());

            Console.Write("Current Reading : ");
            int curr = int.Parse(Console.ReadLine());

            int u = curr - prev;
            double billAmt = CalBill(u);
            double meterRent = GetRent(type);
            double totalAmount = billAmt + meterRent;

            PrintBill(custId, name, addr, phone, email, type,
                prev, curr, u, billAmt, meterRent, totalAmount);
        }

        static double CalBill(int units)
        {
            double amount = 0;

            if (units <= 100)
```

```

        amount = units * 1.5;
    else if (units <= 250)
        amount = 100 * 1.5 + (units - 100) * 2.5;
    else if (units <= 550)
        amount = 100 * 1.5 + 150 * 2.5 + (units - 250) * 4.5;
    else
        amount = 100 * 1.5 + 150 * 2.5 + 300 * 4.5 + (units - 550) * 7.5;

    return amount;
}

static double GetRent(string type)
{
    switch (type)
    {
        case "industrial": return 2500;
        case "business": return 1500;
        case "domestic": return 1000;
        case "agricultural": return 0;
        default: return 0;
    }
}

static void PrintBill(int id, string name, string address, string phone,
                    string email, string type, int prev, int curr,
                    int units, double bill, double rent, double total)
{
    Console.WriteLine();
    Console.WriteLine("-----");
    Console.WriteLine("|          ELECTRICITY BILL RECEIPT          |");
    Console.WriteLine("-----");
    Console.WriteLine("| Customer ID   : {0,-33} |", id);
    Console.WriteLine("| Customer Name  : {0,-33} |", name);
    Console.WriteLine("| Address       : {0,-33} |", address);
    Console.WriteLine("| Phone         : {0,-33} |", phone);
    Console.WriteLine("| Email         : {0,-33} |", email);
    Console.WriteLine("| Connection Type : {0,-33} |", type);
    Console.WriteLine("-----");
    Console.WriteLine("| Previous Reading : {0,-33} |", prev);
    Console.WriteLine("| Current Reading  : {0,-33} |", curr);
    Console.WriteLine("| Units Consumed   : {0,-33} |", units);
    Console.WriteLine("-----");
    Console.WriteLine("| Energy Charges   : ₹ {0,-31} |", bill);
    Console.WriteLine("| Meter Rent       : ₹ {0,-31} |", rent);
    Console.WriteLine("-----");
    Console.WriteLine("| TOTAL AMOUNT    : ₹ {0,-31} |", total);
    Console.WriteLine("-----");
}

```

```
}  
}
```

```
Customer ID : 784  
Customer Name : rohith  
Address : hyderabad  
Phone Number : 4564564561  
Email ID : rohith@gmail.com  
Connection Type : industrial  
Previous Reading : 89890  
Current Reading : 98520  
  
-----+-----  
|                ELECTRICITY BILL RECEIPT                |  
-----+-----  
| Customer ID      : 784                                   |  
| Customer Name    : rohith                               |  
| Address          : hyderabad                             |  
| Phone            : 4564564561                           |  
| Email            : rohith@gmail.com                     |  
| Connection Type  : industrial                           |  
-----+-----  
| Previous Reading : 89890                                 |  
| Current Reading  : 98520                                 |  
| Units Consumed   : 8630                                  |  
-----+-----  
| Energy Charges   : ? 62475                               |  
| Meter Rent       : ? 2500                                |  
-----+-----  
| TOTAL AMOUNT     : ? 64975                               |  
-----+-----  
  
C:\Training\.Net\Exercise4\bin\Debug\net8.0\Exercise4.exe (process 25208) exited with code 0 (0x0).  
Press any key to close this window . . .|
```

Exercise 5:

Boxing is a martial art and combat sport in which two people throw punches at each other, usually with gloved hands. Historically, the goals have been to weaken and knock down the opponent. In December 2025, the World Boxing Council, World Boxing Association and the International Boxing Federation reached an agreement to standardize the names for January 2026 onwards, of the different weight categories under which Boxing matches are conducted.

Input Format:

Input consists of an integer that corresponds to John's weight.

Output Format:

Output consists of a string that corresponds to any of the category names listed above or "Invalid Input".

Print Invalid Input if the weight is negative or if the weight is greater than 120.

namespace Exercise5

```
{  
    internal class Program  
    {  
        static void Main(string[] args)  
        {  
            int wgt = int.Parse(Console.ReadLine());  
  
            if (wgt < 0 || wgt > 120)  
            {  
                Console.WriteLine("Invalid Input");  
            }  
        }  
    }  
}
```

```
    else if (wgt <= 48)
    {
        Console.WriteLine("light fly");
    }
    else if (wgt <= 51)
    {
        Console.WriteLine("fly");
    }
    else if (wgt <= 54)
    {
        Console.WriteLine("bantam");
    }
    else if (wgt <= 57)
    {
        Console.WriteLine("feather");
    }
    else if (wgt <= 60)
    {
        Console.WriteLine("light");
    }
    else if (wgt <= 64)
    {
        Console.WriteLine("light welter");
    }
    else if (wgt <= 69)
    {
        Console.WriteLine("welter");
    }
    else if (wgt <= 75)
    {
        Console.WriteLine("light middle");
    }
    else if (wgt <= 81)
    {
        Console.WriteLine("middle");
    }
    else if (wgt <= 91)
    {
        Console.WriteLine("light heavy");
    }
    else
    {
        Console.WriteLine("heavy");
    }
}
}
```

89

light heavy

C:\Training\Net\Exercise5\bin\Debug\net8.0\Exercise5.exe (process 17128) exited with code 0 (0x0).
Press any key to close this window . . .|